CHAPTER

32

Tooth Discoloration and Bleaching

Esthetics is the science of beauty. Esthetic dentistry not only relates to disfigured teeth but also involves the needs of a normal appearing individual who wishes to look younger, healthier and confident. The perception of color, coupled with desire of the people forms the basis of esthetics. One of the most frequent reasons warranting dental care is discolored anterior teeth. The increasing technical ability of the dental professionals, coupled with public awareness has paved the way for improving the esthetics by lightening the stained teeth. The lightening of stained teeth without disturbing the periodontal environment is the goal of the esthetic operator.

The lightening of color of a tooth using a chemical agent to oxidize the organic pigmentation is referred to as '*bleaching*'. Bleaching is defined as "*the treatment, usually involving an oxidative chemical that alters the light absorbing and/or light reflecting nature of the material structure, thereby increasing its value (whiteness)*".

The terms 'bleaching' and 'tooth whitening' are not exactly synonymous. Whitening restores teeth to their natural tooth color (removal of external stains on the tooth surface) and bleaching makes the teeth lighter than their natural color (modify colored substances within the tooth).

Bleaching or tooth whitening dates back to 18th century when chloride of lime was used as bleaching agent. Different authors, over the years, tried various materials and techniques to aid in bleaching. In early nineteenth century, hydrogen peroxide alone and in combination with other materials, was used as bleaching agent. Lasers and other modalities have also been tried to bleach the discolored teeth.

TOOTH DISCOLORATION AND STAINING

The tooth discoloration and staining are multifactorial. A thorough knowledge of the etiology of tooth staining is essential in order to make a correct diagnosis. The factors and extent of discoloration influence the treatment options and also determine the prognosis. The vitality of tooth also affects the treatment protocol and timings.

Etiology

The etiology of tooth discoloration is broadly divided into three categories, viz. intrinsic, extrinsic and internalized stains.

A. *Intrinsic Discoloration*

Intrinsic discoloration implies color changes within the tooth tissues.

The factors causing intrinsic discoloration are classified as local and systemic factors.

a. *Local factors*

i. *Trauma:* Trauma leads to pulpal hemorrhage, causing grayish discoloration of the teeth. The obliteration of pulp canal may add to dark yellow discoloration of the teeth.

ii. *Pulpal hemorrhagic products:* Pulp necrosis leads to the production of hemorrhagic products, which enter dentinal tubules and cause discoloration.

iii. *Aging:* Aging changes enamel and dentin structure, which eventually lead to discoloration.

iv. *Root resorption:* Root resorption, especially at a later stage, discolor the tooth.

b. *Systemic factors*

i. *Fluorosis:* It has been established that high concentration of fluorides (more than 4.0 ppm) causes moderate to severe discoloration; more prevalent in incisors, followed by premolars and molars. Fluorosis is categorized as mild, moderate and severe. Mild fluorosis exhibits brown pigmentation on a smooth surface, moderate shows flecks on the enamel surface along with staining, and severe form shows pigmentation with pitted enamel surfaces. The severe form of fluorosis may not respond to bleaching.

ii. *Tetracycline staining:* Tetracycline is a widely used antibiotic in routine infections. Discoloration due to tetracycline ingestion has been confirmed by various investigators. It is established that the dose of tetracycline is more important than the duration of intake of tetracyclines. The different varieties of tetracycline produce yellow-to-yellowish-brown stains; however, doxycycline does not produce any stain.
iii. *Amelogenesis imperfecta:* Amelogenesis imperfecta, the abnormal formation of the enamel, is of three types, viz. hypoplastic, hypomineralization and hypomaturation. All three types produce discoloration of teeth.
iv. *Dentinogenesis imperfecta:* Dentinogenesis imperfecta is an autosomal dominant developmental disturbance of the dentin. It leads to brownish-violet to yellowish-brown color of the dentin with a translucent hue.
v. *Dentin dysplasia:* Dentin dysplasia is a rare disturbance of dentin formation; may lead to slight amber translucency of teeth.
vi. *Congenital hyperbilirubinemia:* High level of bilirubin in the blood; may cause yellow green discoloration of teeth.
vii. *Congenital erythropoietic porphyria:* An inborn error of porphyrin-haem synthesis (inherited as an autosomal recessive trait); may lead to red brown/purplish-red discoloration of the teeth.
viii. *Alkaptonuria:* An inherited genetic disorder of phenylalanine and tyrosine metabolism; may lead to brown discoloration of teeth.
ix. *General diseases:* Diseases of genetic/idiopathic origin, such as, epidermolysis bullosa, cleidocranial dysostosis, osteogenesis imperfecta, osteopetrosis, Morquio's disease, prenatal/congenital syphilis, trophic disturbances like infantile tetany, rickets, vitamin C deficiency, etc.

A few authors have categorized causes of intrinsic stains as pre-eruptive and post-eruptive, as follows:

Pre-eruptive

- Tetracycline
- Fluorosis
- Trauma
- Genetic (amelogenesis imperfecta, hyperbilirubinemia)

Post-eruptive

- Aging
- Root resorption
- Pulp necrosis
- Intrapulpal hemorrhage
- Pulp tissue remnants
- Materials (irrigants, sealers, etc.)

B. Extrinsic Discoloration

Extrinsic discoloration implies color changes on the external tooth surface. Extrinsic discoloration have been classified by various authors depending upon different criteria:

a. *Classification according to the cause*

- Staining, caused by chromogenic microorganisms derived from dietary sources; other factors may also be responsible.
- Indirect staining is associated with certain antiseptics and metal salts.

b. *Nathoo's classification*

- Type 1 (N1): Chromogens bind to tooth surface and cause discoloration. The color of the chromogen is similar to that of dental stains caused by tea, coffee, wine and chromogenic bacteria.
- Type 2 (N2): Chromogen changes color after binding with teeth. The stain darkens with time.
- Type 3 (N3): Pre-chromogen (colorless material) binds to tooth and undergoes chemical reaction, causing the staining.

c. *Non-metallic stains*

- Due to tea, coffee, blackcurrant juice, cola drinks, etc.
- Due to chemicals, like Chlorhexidine, etc.

d. *Metallic stains*

- Black staining seen in people using iron supplement and in iron foundry workers.
- Copper causes green stain after rinsing with copper salts.
- Potassium permanganate produces violet to black color.
- Silver nitrate salt causes gray discoloration.
- Stannous fluoride causes golden-brown discoloration.

C. Internalized Discoloration

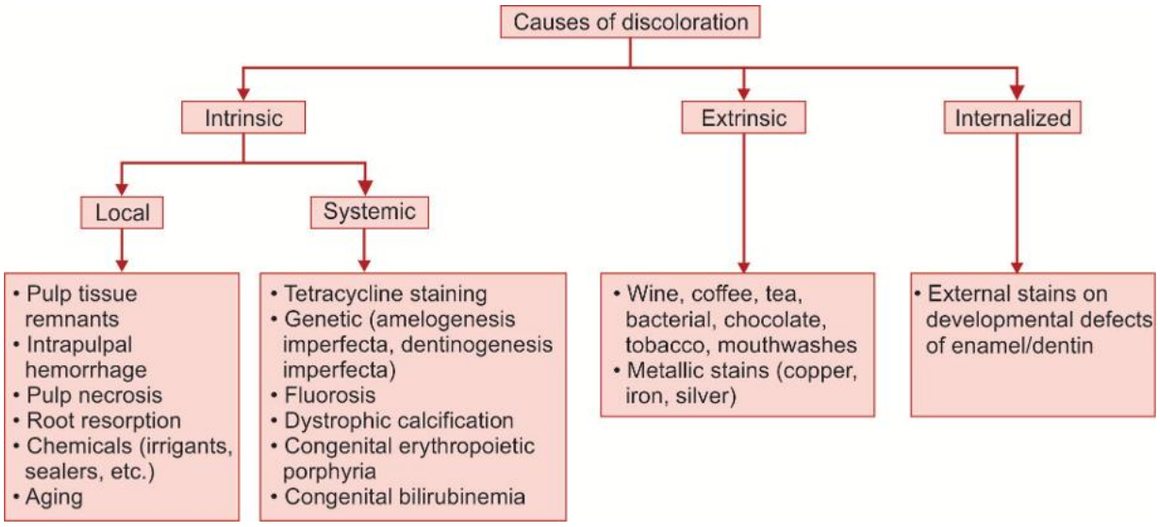
Internalized discoloration occurs due to incorporation of extrinsic stains within the tooth substance following some dental pathology. It occurs in enamel defects and in the porous surface of exposed dentin; for example, in developmental defects, like fluorosis, hypoplasia and enamel hypocalcification. These types of defects can also be seen in tooth wear, gingival recession and caries, etc.

The causes of discoloration are summarized in Flowchart 32.1.

BLEACHING OF TEETH

Bleaching implies degradation of high molecular weight, complex organic molecules that reflect a specific wavelength of light, which is responsible for color of the stain. The resulting degradation products

Flowchart 32.1: Causes of discoloration



are of lower molecular weight that reflect less light, resulting in reduction or elimination of discoloration.

Indications

- Persons dissatisfied with the original color of their otherwise sound teeth.
- Fluorosis stains (not very deep).
- Minor tetracycline stains
- Discolored teeth that have darkened from trauma or post-endodontic therapy.
- As retreatment of bleaching.
- Discolored teeth, even though considered for the placement of porcelain/esthetic veneer.
- Teeth affected with dentinogenesis imperfecta.
- To lighten natural teeth to match existing ceramic crowns, fixed partial dentures, etc.

Contraindications

- Severe discoloration from amalgam corrosion
- Deep tetracycline stains
- Severe fluorosis
- Extensive restorations or caries
- Inherent sensitivity of the patient to bleaching agents

Bleaching Agents

The bleaching agents contain active and inactive ingredients.

1. *Active Ingredients*

The active ingredients include:

i. *Chlorine dioxide:* 0.5% chlorine dioxide when applied to enamel surface for 20 minutes, effectively bleach the tooth surfaces. However, its use may lead to roughness on enamel surface and decreased lustre. Such teeth are more prone to re-staining and usually develop sensitivity.

ii. *Hydrogen peroxide:* 30–35% hydrogen peroxide is a strong oxidizing agent that ionizes to produce different types of free radicals, such as hydroxyl ions (OH^-) and perhydroxyl ions (OOH^-).

iii. *Carbamide peroxide:* Carbamide peroxide is hydrogen peroxide combined with urea. 10–35% concentration of carbamide peroxide is being used

Bleaching agents

- *Hydrogen peroxide (30–35%):* It is a strong oxidizing agent. Ionization of H_2O_2 produces different types of free radicals, such as hydroxyl ions (OH^-) and perhydroxyl ions (OOH^-).
- *Hydrochloric acid (18–36%):* The hydrochloric acid causes decalcification of tooth substance along with removal of stain.
- Superoxol and pyrozone (H_2O_2 + ether)
- Sodium perborate and sodium percarbonate
- *McInnes solution:* 5 parts of 36% HCl (increases the penetration of solution) + 5 parts 30% H_2O_2 (bleaches the enamel by process of oxidation) + 1 part of 0.2% anesthetic ether (removes the surface debris).
- *Modified McInnes bleaching solution:* 30% H_2O_2 and 20% NaOH in 1:1 ratio along with 0.2% ether, introduced to overcome the deleterious effects of hydrochloric acid.
- *Self-activating bleaching agent:* 35% H_2O_2 1.4 mL, CaO 0.12 gm, aerosil 0.32, 1.48 or 0.64 gm (to alter/control the viscosity).
- *Other bleaching agents:* For example, Fuji Hilite dual cure material (It is green-blue in color and on activation loses oxygen and color of particle changes to white). It can be used in-office or at-home.

It has been established that hydrogen peroxide above 15% concentration does not increase bleaching effectiveness; may lead to alteration in enamel hardness and surface morphology.

for bleaching purpose (10% carbamide peroxide is equal to 3.0% hydrogen peroxide and 22% carbamide peroxide is equal to 7.5% hydrogen peroxide). Higher amount of peroxide is more effective; however, associated with increased risk of sensitivity. A few pH buffered products are available, which can minimize the side effects.

iv. *Hydrochloric acid:* The hydrochloric acid (18–36%) causes decalcification of tooth substance along with removal of stain.

v. *Titanium dioxide nanoparticle doped with nitrogen* with less concentration of hydrogen peroxide (3.0–10.0%) is also used as bleaching agent. The combination improves the biocompatibility of the solution reducing sensitivity during and after the procedure. A few studies have confirmed the clinical efficacy of this newer bleaching agent with 3.0–6.0% hydrogen peroxide.

vi. *Other bleaching agents:* The whitening toothpastes and tooth whitening gels mainly contain Calprox-baking soda-sodium bicarbonate (mild abrasive, dissolves protein pellicle layer and whiten teeth); hydrated silica-silicone dioxide (mild abrasive, can be combined with calcium carbonate to increase effectiveness); sodium tripolyphosphate (detergent, removes stains) and polyvinylpyrrolidone (do not allow new stains adhering to tooth).

2. Inactive Ingredients

The inactive ingredients include:

i. *Preservatives:* Sodium benzoate is commonly used as preservative agent. It prevents bacterial growth and also accelerates the breakdown of hydrogen peroxide.

ii. *Thickening agents:* Carbopol (carboxypolymethylene), in a concentration of 0.5 to 1.5%, is commonly used thickening agent. It increases viscosity of the bleaching agent and also increases active oxygen releasing time of the bleaching material.

iii. *Surfactants in bleaching agent* act as surface wetting agent, facilitating diffusion of hydrogen peroxide across the tooth surface.

iv. *Glycerine and propylene glycol,* added as carrier, enhance viscosity and facilitate uniform distribution of bleaching agent.

v. *Flavoring agents*, such as peppermint, wintergreen, spearmint, etc., are added to improve taste and patient's acceptance.

vi. *Remineralizing agents*, such as calcium phosphate-casein phosphopeptide, are added to reduce sensitivity and also to get lustrous shine to the teeth.

vii. *Potassium nitrate and fluorides* are added to decrease post-treatment sensitivity.

Additional Materials used during Bleaching Procedure

The following materials are commonly used along with the bleaching agents:

a. ***Orabase paste:*** Orabase paste is used to protect gingiva and oral mucosa during bleaching. It is composed of gelatin, pectin and sodium carboxymethylcellulose in plastibase (plasticized hydrocarbon gel). Orabase paste adheres tenaciously and remains in intimate contact with mucous membrane, protecting the affected area in the oral cavity against irritation from chewing, swallowing, etc. It acts as an invisible bandage. A thin film of orabase paste is effective.

b. ***Gingival dam:*** The gingival dam is the polymerizable isolation barriers, which is used during bleaching process. The barrier material consists of:

- Monomers
- Curing agents (0.01–2%)
- Polymerization strength reducers (1–30%)
- Tissue adherence accentuators (0.01–9%)
- Reflective materials (1–50%)

The polymerizable isolation barrier consists of at least one monomer, one curing agent for curing the monomer and one of three preferred additives, i.e. the organic polymerization strength reducer, the tissue adherence accentuator, and the reflective material. The polymerizable isolation barrier material is preferably stored at or below room temperature. It is stable enough until activated by suitable light radiant energy. The barrier material is made in a paste or gel form that can be expressed from a conventional syringe. The barrier material is available in the form of emulsion, dispersion/suspension solution, depending upon selection of a preferred application.

c. ***Diamond polishing paste:*** The diamond polishing paste incorporates 3.0 μ diamond abrasive particles in a strand of fiber. These fibers form a three dimensional open weave design. This paste provides finishing and polishing in one step.

d. ***First aid kit:*** It contains antioxidants, such as vitamin E in liquid or capsule form and aloe-vera gel. A single spill of a droplet of hydrogen peroxide blanches and burns the gingival tissue. Vitamin E oil quickly relieves the symptoms within one minute.

BLEACHING MODALITIES

The bleaching modalities are divided into vital and non-vital tooth bleaching.

I. Vital tooth bleaching
   A. In-office bleaching

a. Thermocatalytic bleaching
b. Power bleaching
c. Non-thermocatalytic bleaching
d. Microabrasion-assisted bleaching

B. Home bleaching (nightguard bleaching)

II. Non-vital tooth bleaching
a. Thermocatalytic technique
b. Walking bleach technique
c. Modified walking bleach technique
d. Combination bleaching technique (inside/outside bleaching)

III. Laser-assisted bleaching
IV. Bleaching with non-thermal atmospheric plasma
V. Over the counter products
VI. Advanced techniques

I. VITAL TOOTH BLEACHING

Vital bleaching technique requires use of materials that do not endanger the tooth structure and the soft tissue.

Methods of Vital Bleaching

Two basic methods of vital bleaching are:

A In-office bleaching
a. Thermocatalytic bleaching
b. Power bleaching
c. Non-thermocatalytic bleaching
d. Microabrasion-assisted bleaching

B. Home bleaching (nightguard vital bleaching)

A. In-office Bleaching

Preparation of the Patient

- Shade of the patient's existing dentition is recorded with a standardized shade guide tab.
- Photograph of patient's teeth is taken for records.
- Thorough prophylaxis is performed.
- Patient is draped with a protective drape and is made to wear protective eye glasses.

Advantages

- Treatment is under control of the operator.
- Potential for early results.
- Soft tissue is protected during bleaching.

Disadvantages

- Unknown duration of the treatment.
- Discomfort of rubber dam.
- Adverse effects of thermocatalytic temperature rise on the pulp.
- Resultant post-treatment sensitivity.
- If etching is performed with bleaching, polishing is required after each visit, resulting in small amount of enamel loss.
- Local anesthesia is not required.
- Teeth are isolated with heavy gauge rubber dam.
- Before rubber dam application, oraseal (a light cure resin) or orabase paste is applied to protect the labial and lingual soft tissue.
- Vaseline is applied to the patient's lips before mounting the rubber dam frame.
- Wet gauze is placed over the patient's lips to prevent thermal trauma.

a. Thermocatalytic Bleaching

The thermocatalytic bleaching involves the use of heat alone or heat and light both. The units commonly used are:

i. *Heat-light unit:* The bleaching light provides high intensity light and heat, required to activate bleaching agents. A narrow beam of light is concentrated in one section of the oral cavity at a distance of approximately 13–15 inches. Calibrated rheostat controls the amount of light and heat.
ii. *Heat unit:* Heat unit with accurate temperature control is used to activate the bleaching agent.

The recommended temperature is 46°C to 60°C. Bleaching agent is applied uniformly and treated for 30 minutes with light/light-heat.

After removing the heat source, tooth is allowed to cool to avoid sudden temperature change that can be deleterious to the pulp. After five minutes, tooth is washed with warm water. Bleaching appointments are scheduled 2–4 weeks apart. A minimum of three treatments are usually sufficient; however, extra treatments can be carried out (not more than ten). Color is checked one week after the third treatment.

Sources of heat

- Photoflood lamp
- Polymerization light
- Spirit lamp
- Commercial bleaching units
- Light-heat lamp
- Lasers (argon and diode)

Some of the commercial systems utilizing different activating light sources are:

i. *Quartz-tungsten-halogen curing light:* The standard curing light provides heat to simulate the chemical reaction by activating light sensitive chemicals in the bleaching agent. Each application is used for 40–60 seconds.
ii. *Plasma arc:* It provides intensity of light similar to or slightly higher than halogen curing temperature. An application of four seconds leads to an intrapulpal temperature rise of 2.2°C. The allowable time of application is 30 seconds per tooth.

iii. *Rembrandt tooth whitening system:* This system includes the use of a plasma arc light (named Rembrandt Sapphire). The bleaching with a Rembrandt Whitening Crystal facilitates the operator to treat both the arches simultaneously. The wavelength of light emitted from this source is in the range of 400–525 nm (blue-green color range).

iv. *Whitening accelerator:* This system uses high intensity blue light at a wavelength of 480–520 nm. The light is filtered to remove infrared and ultraviolet light. It is used with special whitening formula; half hour procedure can oxidize 16 or even more teeth.

v. *Zoom! Teeth whitening system:* This system includes the use of mercury metal halide light with a wavelength of 300–450 nm (violet color range). Zoom! Light unit has an infrared filter, which filters these radiations, thereby minimizing the amount of heat generated at the surface of tooth during the whitening treatment.

Procedures

- Using an eye dropper, a small amount of bleaching solution (30–35% H_2O_2) is placed into a dappen dish.
- Teeth are isolated, preferably with rubber dam.
- The cotton or gauze piece, held in cotton pliers, is dipped into the saturated bleaching solution and carried onto the tooth surface.
- With a small plastic instrument, the cotton/gauze is shaped to cover the entire surface.
- The selected heating unit is positioned.
- When the photo lamp is used, the solution is applied every 4 to 5 minutes; when heat unit is used, the solution is applied after every heating cycle.
- After removing the gauze/cotton, the teeth and the rubber dam are washed with warm water and a high vacuum aspirator.
- The rubber dam is stretched apically and labially to gain access to the interproximal areas. The clamps are removed and the rubber dam sheet is gently teased off from the teeth.
- Wet, warm gauze is used to remove the excess orabase/oraseal.
- The residues on teeth are neutralized with a mild warm solution of ½ teaspoon salt, ½ cup warm water and ½ teaspoon sodium bicarbonate.
- Instruct the patient to avoid coffee, tea and cola drinks for two weeks.

Figure 31.1A to C depicts vital (thermocatalytic) bleaching on maxillary central incisors.

b. *Power Bleaching*

In this technique, high intensity light, conventionally used as a heat source, is replaced with halogen light,

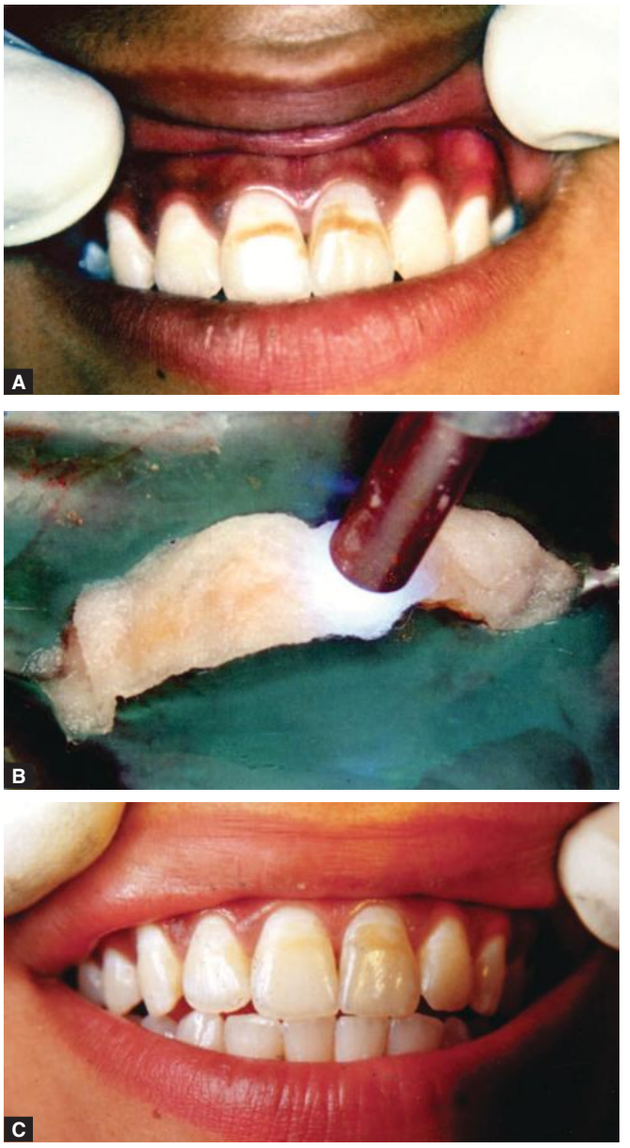


Fig. 32.1: Vital bleaching (thermocatalytic technique): **(A)** Preoperative; **(B)** Application of bleaching agent; **(C)** Post-operative

Advantages
• Produces immediate results. • Avoiding problems with home bleaching procedures, such as: – Wearing trays that cause patients to gag. – Distaste for home bleaching gel.

Disadvantages
• Caustic nature of 30–35% H_2O_2 makes isolation and protection mandatory. • Dehydration of teeth may occur, thereby, giving a false lighter shade immediately post-treatment. • Expensive

plasma arc lamps, LED lights, Xe-halogen lights and lasers. There are specific situations, such as single tooth bleaching within an arch, or even small areas on teeth that need to be lightened.

c. Non-Thermocatalytic Bleaching

The non-thermocatalytic technique does not utilize heat sources.

After rubber dam application, paste is applied on the teeth for 5 minutes and re-application is carried out as required followed by copious irrigation of warm water and polishing (Fig. 32.2A to E).

d. Microabrasion-Assisted Bleaching

Microabrasion is a procedure involving the surface dissolution of the stains of enamel by an acid (preferably 18% HCl) along with abrasives (pumice powder). Microabrasion is indicated for improvement of particular tooth color. This is also used in cases where routine bleaching is not effective.

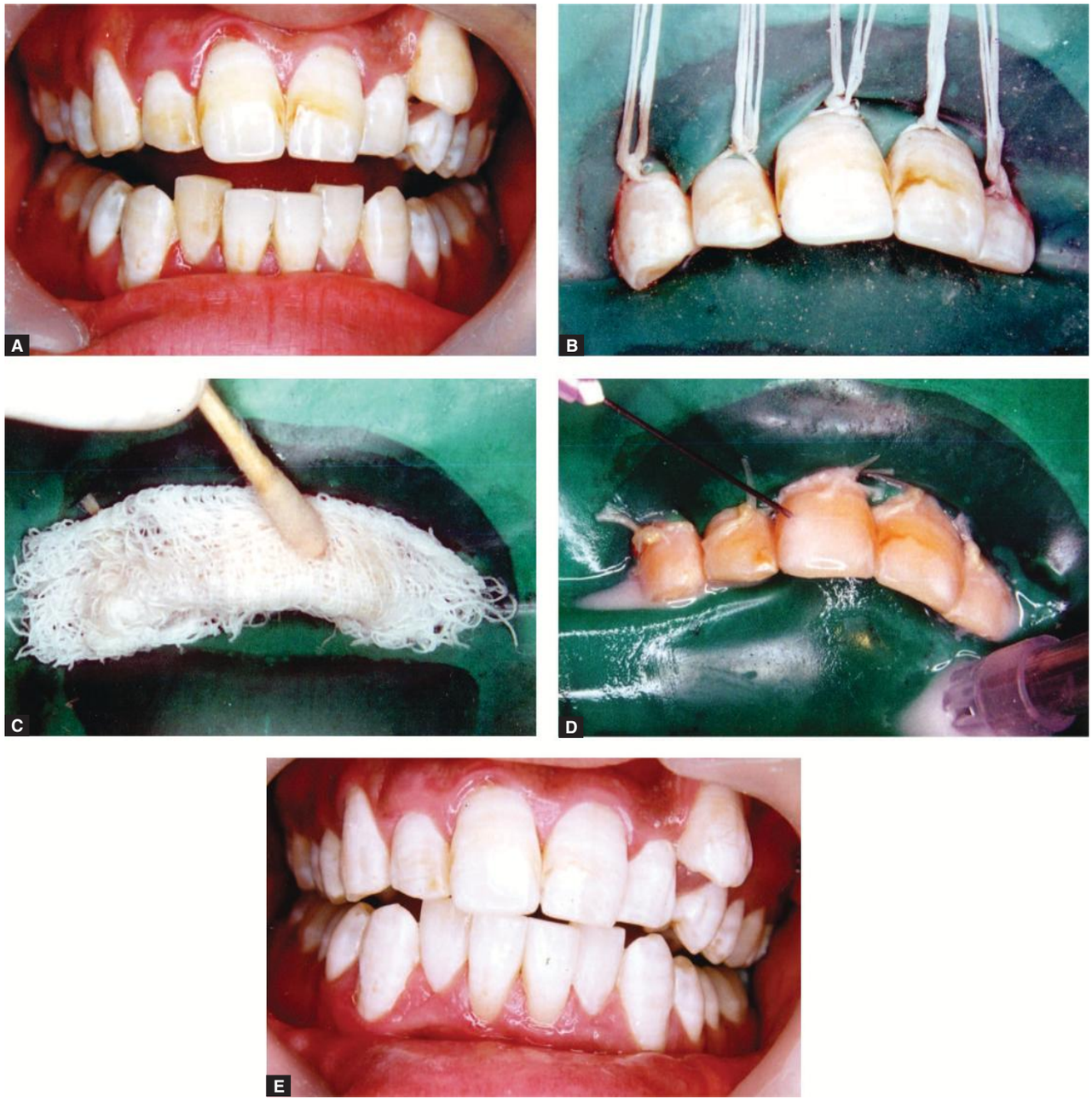


Fig. 32.2: Vital bleaching (non-thermocatalytic technique): **(A)** Preoperative; **(B)** Rubber dam application; **(C)** Application of bleaching agent; **(D)** Irrigation; **(E)** Postoperative

Advantages

- Helps to remove superficial stains and discoloration.
- Minimal patient discomfort and operator difficulty.
- Treated teeth display a smooth texture and shine.

Disadvantages

- Microabrasion removes enamel layer.
- Only effective on those stains confined to outer layer of enamel, i.e. not effective for deeper stains
- Since it removes enamel layer, teeth sometimes appear more yellow after treatment.

Indications

- Stains limited to superficial enamel surface
- Enamel fluorosis (initial stages)
- Discoloration due to enamel hypo- or hypermineralization.
- Small decalcified lesions, may be due to orthodontic bands.

Contraindications

- Deep stains
- Stains of amelogenesis imperfecta, dentinogenesis imperfecta and tetracycline stains
- Hypoplastic lesion
- Decalcification areas leading to caries

Technique

- Selected tooth/teeth are cleaned with prophylaxis paste using rubber cups at slow speed.
- Teeth are isolated using rubber dam.
- Petroleum jelly is applied to adjacent soft tissue.
- Microabrasive agent is applied to the tooth surface for 60 seconds, followed by rinsing.
- Process is repeated till the requisite result is achieved.
- Topical fluoride can be applied after the treatment.
- Patient's consent is necessary for the final color achieved; if not satisfied, the process is repeated.

Figure 32.3A to D depicts microabrasion-assisted bleaching on maxillary incisors.

B. Home Bleaching (Nightguard Bleaching)

Earlier, nightguard bleaching utilized 10% carbamide peroxide, delivered in a custom fitting tray. The patient is asked to keep the tray alongwith the medicament over the teeth during night time (Fig. 32.4A, B).

Bleaching agent is flown onto the tooth surface, (carbamide peroxide being unstable dissociates into water, urea and/or oxygen). The carbamide peroxide dissociates into H_2O_2 and urea, in different concentrations, depending upon the concentration of the carbamide peroxide. The H_2O_2 further dissociates into water and oxygen. The oxygen radicals cause oxidation of pigmentation in the tooth. The urea degrades into ammonia and carbon dioxide. An agent, carbopol is added to prolong the release of oxygen, counter carbon dioxide released by urea and improve tissue adherence.

Advantages

- Patient convenience
- Cost-effective

Disadvantages

- Patient compliance is mandatory
- Difficult for operator to monitor the amount of time the trays are used as well as the color change.

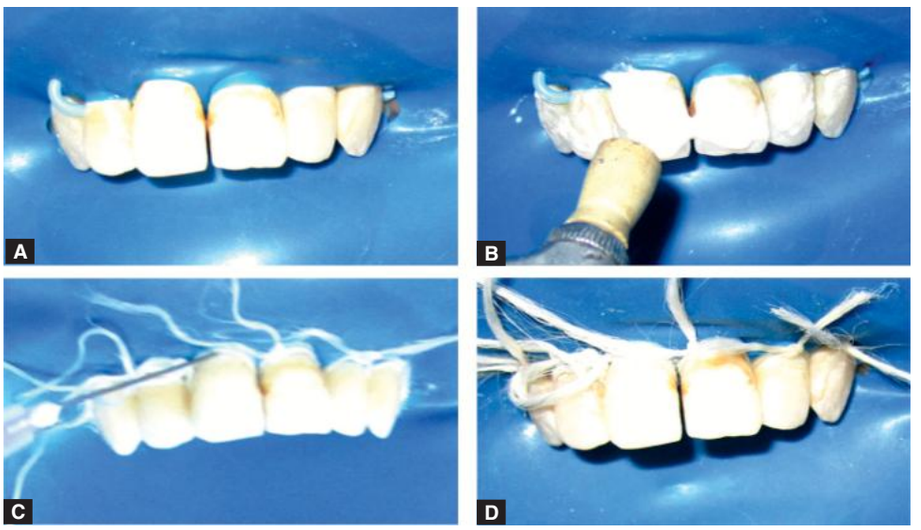


Fig. 32.3: Microabrasion-assisted bleaching: **(A)** Preoperative; **(B)** Polishing the tooth surface; **(C)** Cleaning/irrigating after the procedure; **(D)** Postoperative



Fig. 32.4A: Tray and syringe



Fig. 32.4B: Bleaching trays

Commercial Preparations (Chemical Agent)

- 10% carbamide peroxide with carbopol (proxigel, ultralite, etc.)
- 10% carbamide peroxide without carbopol; fast oxygen releasing, (Glyoxide, Dentalite, etc.)
- 15% carbamide peroxide, (Nu smile).
- 1.0–10% H_2O_2 (Peroxyl, Brite smile, etc.)

Procedure

The three regimens for applying the whitening solutions are:

i. Sleeping with the nightguard tray filled with the bleaching solution (Solution is changed each night).
ii. Wearing the loaded nightguard tray during the day while changing the solution every 1.5–2 hours.
iii. Polyethylene strips impregnated with 5.25% of H_2O_2 is also used without tray.

The procedure usually requires three appointments.

First appointment: The initial shade examination and evaluation of the adjacent structures is carried out before the treatment. The cast is formed after taking impressions with suitable materials and the tray material is selected. Usually 0.040 inch or 0.035 inch ethyl vinyl acetate is used for fabricating the tray; however, 0.020 inch polypropylene has also been used.

After selection of tray material, the cast is placed on the suction grid. The tray material is placed in the retainer frame on the vacuum forming unit. When the tray material is sufficiently softened by heating, it sags approximately one inch; the heated sheet is pulled down and adapted slowly and gently.

The tray material is allowed to cool on the cast. The nightguard is trimmed by scalloping the tray in a smooth curve until only about 2.0 mm of tissue apical to gingival crest is covered facially and lingually.

Variations in tray fabrication

i. Round bur no. ½ is used to cut groove along the gingival tooth margin on the cast.
ii. A gel trench is built into the tray by stretching a rectangular rubber band (2.0 mm × 2.0 mm) around the crest, centred on the clinical crowns of the teeth. This trench functions as blocked out reservoir.
iii. Nightguard is also prepared using reservoir space:
   - Reservoirs are created using a viscous light cure composite gel.
   - Viscous composite spacer gel is applied to the selected teeth on the cast.
   - Only the facial surface of the teeth is painted 1.0 mm from the gingival margin with a thickness of 0.5 mm.
   - The spacer gel is not allowed to flow onto the incisal edge or occlusal surface.
   - A light curing oven or the clinical light curing wand can be used for curing the spacer.
   - The surface is washed or the oxygen inhibited resin layer is wiped away from the spacer.
   - Tray material is selected and nightguard constructed in a vacuum forming machine.
   - Space of spacer gel in the bleaching trays forms the reservoir.

Advantages
• Provides space for the bleaching agent. • Allows seating of tray when more viscous solution is used. • Minimizes tooth sensitivity by avoiding the pinching effect of a light tray. • Allows the transfer of bleaching gel in area of higher concentration.

Disadvantages
• Increased cost of fabrication and time. • Decreased tray retention. • Gel leakage and irritation of gingiva.

The tray is also formed using foam liners, which are placed on the selected area of the cast. The trays are prepared in a vacuum forming machine.

Second appointment: The second appointment involves inserting and fitting the nightguard. The bleaching material is applied into the nightguard as follows:

- Two to three drops of bleaching material is placed into the area of each tooth to be bleached.
- After inserting the nightguard, the excess material is wiped out.

Advantages of foam liner
• Soft on patients' gingival tissues • 50% reduction in bleaching time

Disadvantages of foam liner
• Esthetically less pleasing • Occlusal discrepancies are common • Does not help in retention

- Patient is instructed not to drink or rinse during treatment.
- The bleaching solution is replaced every one and half to two hours during daytime regimen.
- A single application of bleaching material at bedtime is indicated, if worn at night.
- The daytime regimen requires one to three weeks; whereas, 4 to 6 weeks are required for night- time bleaching.

Third appointment: This appointment includes post-operative photographs, patient's satisfaction and consent; a decision whether future bleaching or restorative treatment is required (Fig. 32.5A, B).

The wearing time of bleaching agent in the tray is being measured and recorded. A microsensor (Thermon Microelectronic system) is embedded into the tray. The sensor records the wearing time of the tray, as well as wearing off of carbamide peroxide with time. The process help the clinican and the patient to modify the treatment accordingly.

II. NON-VITAL TOOTH BLEACHING

The non-vital teeth are discolored because of pulpal necrosis and related complications. Many a times, to achieve better results in deeply stained teeth (tetracycline stains), intentional root canal treatment is carried out so as to bleach the tooth from both inside and outside. Calcified root canals without any root canal treatment are also candidates for non-vital bleaching. However, this modality is contraindicated in teeth, which are discolored due to corrosion products and/or large restorations.

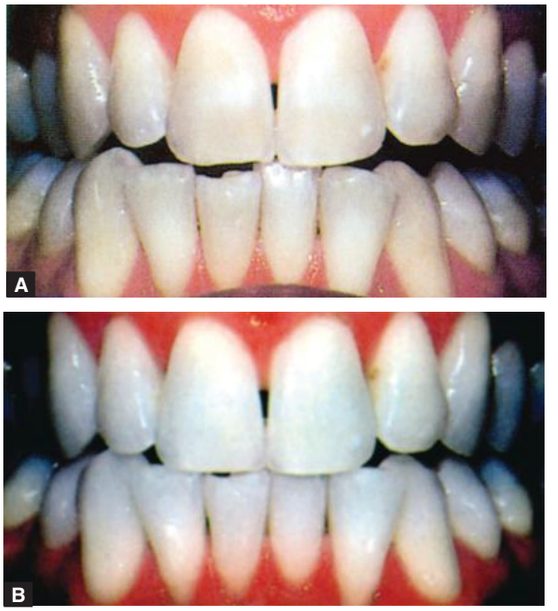


Fig. 32.5: Nightguard vital bleaching: **(A)** Preoperative, **(B)** Postoperative

Methods of Non-Vital Bleaching

The basic methods of non-vital bleaching are:

a. Thermocatalytic technique
b. Walking bleach technique
c. Modified walking bleach technique
d. Combination bleaching technique (inside/outside bleaching)

Preparation of the Tooth

- The restoration, if any, is removed from the coronal opening and from within the pulp chamber.
- Approximately 2.0–3.0 mm of root canal restorative material is also removed in an apical direction beyond cementoenamel junction (Fig. 32.6A).
- Pulp chamber is cleaned and superficial dentin, if present, is removed.

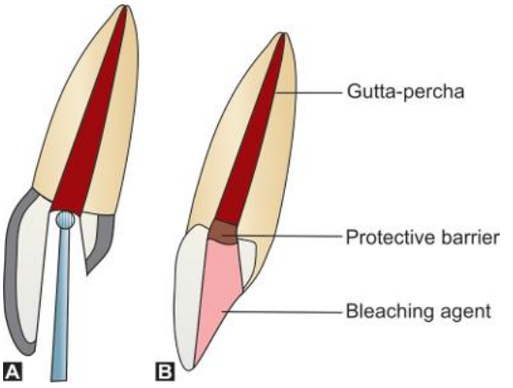


Fig. 32.6: Preparation of tooth for non-vital bleaching: **(A)** Removal of gutta-percha and coronal filling; **(B)** Placement of barrier material

- The tooth is washed with 3.0% H_2O_2 solution, rinsed with water and dried.
- 0.5–1.0 mm thick calcium hydroxide/glass-ionomer cement plug is sealed in direct contact with the obturation material (act as mechanical barrier) (Fig. 32.6B).
- The pulp chamber is again washed and kept prepared to place the bleaching agent.

a. Thermocatalytic Technique

- Cotton pellet is placed against labial aspect of the tooth.
- Bleaching agent (superoxol and sodium perborate) separately or a combination, soaked in cotton pellet, is placed in the root canal.
- Solution is heated with a bleaching wand at low to medium setting. A dry endodontic spreader or heated spatula can also be used. Light heating units are also employed.
- The process is repeated as required with the total treatment time per appointment ranging from not more than 20–30 minutes along with remoistening of the cotton pellets at regular intervals.
- Tooth is rinsed with water. A fresh, dry cotton pellet is placed in the pulp chamber and sealed with temporary sealing agent.
- Patient is recalled after one to three weeks.
- Final restoration of the cavity is carried out preferably with composite resin, the shade esthetically matching with the tooth. *(It is recommended that acetone-based adhesive systems be used because these have the potential to reverse the adverse effects of bleaching on enamel bond strengths).*

Figure 32.7A to G depicts non-vital (thermocatalytic) bleaching on maxillary left central incisor.

Thermocatalytic Technique: Variations

i. A mixture of one part 95% ethyl alcohol and two parts chloroform is used to desiccate the exposed enamel and dentin for two minutes. This is followed by placement of cotton pellet saturated with pyrozone in the access cavity and on the labial and lingual surfaces of the teeth. A thermocatalytic source (a photoflood lamp) is used for about 20 minutes. After this, a dressing of 30% H_2O_2 is placed in pulp chamber and sealed with appropriate material.
ii. A soldering metal encased in a case whose temperature can be regulated has also been used. The bleaching solution used is four part 30% H_2O_2 and one part ethyl ether.
iii. The dentinal tubules are dehydrated with 90% ethyl alcohol. After this, a vulcanite rubber cone is pressed for 1–2 minutes over the cotton saturated with superoxol or pyrozone into the pulp chamber. The bleaching agent is activated with a light source (lamp) for a period of 30–40 minutes.
iv. A mixture of 95% ethyl alcohol and chloroform (3:1) is used for dehydrating the dentin and enamel. A 30% H_2O_2 saturated cotton pellet is placed into the access cavity. This bleaching agent is activated with ultraviolet light for two minutes. After the desirable results, the cavity is filled with gutta percha within 1.5 mm of its margins followed by composite resin.
v. A combination of 35% H_2O_2, manganese sulfate as photo activator, and iron sulfate as chemical activator has also been used.

b. Walking Bleach Technique

It employs a bleaching regimen, which involves placement of the bleaching agents in the pulp chamber over an extended period of time ranging from 24–48 hours to 7–10 days. The pulp chamber is sealed with appropriate materials. The commonly employed agents are superoxol, sodium perborate and their combinations.

Procedure

- A thick paste of sodium perborate and superoxol (35% H_2O_2) is placed into the pulp chamber.
- The lingual access is sealed with a suitable material. It should also create pressure, as pressure in the pulp chamber is required for bleaching. The commonly used sealing agents are fast setting zinc oxide eugenol, glass-ionomer cement, etc.

Figure 32.8A to C depicts walking bleach technique on maxillary left central incisor.

c. Modified Walking Bleach Technique

i. Aldecoa and Mayordomo (1992) modified the technique for severe tetracycline discoloration. A mixture of 10% carbamide peroxide and sodium perborate is placed in the pulp chamber for 4 to 6 weeks.
ii. Liebenberg (1997) administered intracoronal 10% carbamide peroxide gel in a splint constructed from 0.02"–0.03" polypropylene coping material. The splint is used to retain the bleaching agent and to prevent ingress of debris into the access cavity.

d. Combination Bleaching Technique (Inside/Outside Technique)

This technique involves the combined use of thermocatalytic non-vital bleaching technique followed by walking bleach regimen using a mixture

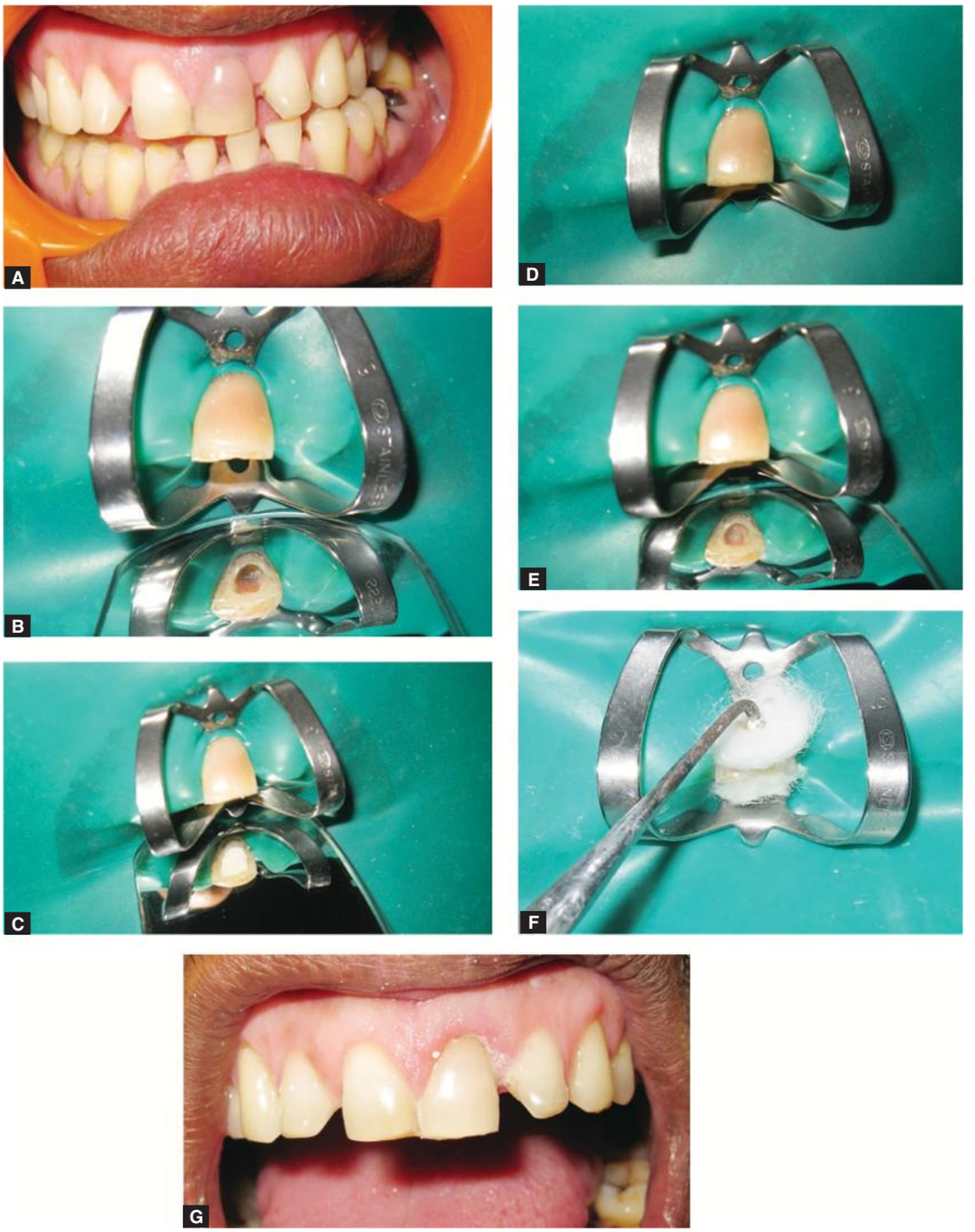


Fig. 32.7: Thermocatalytic technique (non-vital bleaching): **(A)** Preoperative; **(B)** Isolation with rubber dam; **(C)** Removal of coronal restoration; **(D)** Sealing of cervical area with GIC; **(E)** Placement of bleaching agent; **(F)** Application of heat; **(G)** Post-operative

of sodium perborate and superoxol. It provides synergistic effect and is significantly effective (Fig. 32.9).

III. LASER-ASSISTED TOOTH WHITENING

Laser tooth whitening utilizes 30–35% H_2O_2, which is usually applicable in routine bleaching. Laser whitening gel has a unique mix of thermal absorption crystals integrated into gel of highly processed fumed silica and 35% H_2O_2.

Bleaching gel is applied and is activated by high intensity light source or plasma arc light. Crystals in gel absorb thermal energy from light, allowing better dissociation of oxygen and easy penetration into the enamel matrix, thus increasing the lighting effect on teeth.

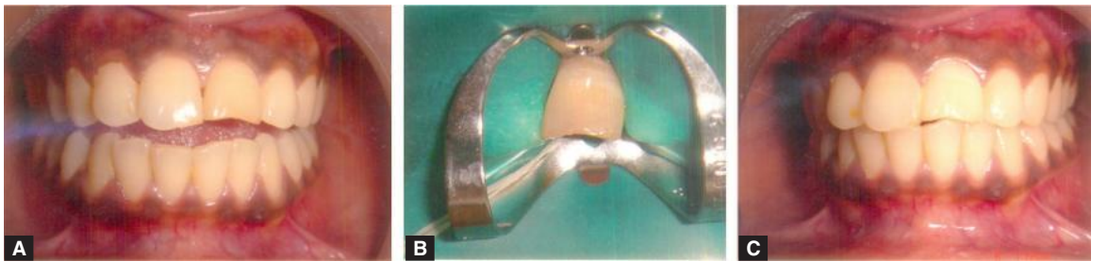


Fig. 32.8: Walking bleach technique: **(A)** Preoperative; **(B)** Placement of bleaching agent in the coronal access area; **(C)** Postoperative

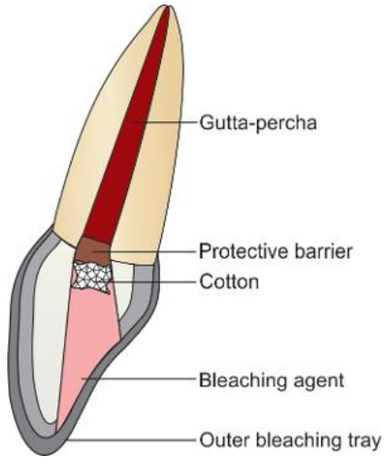


Fig. 32.9: Inside/outside bleaching (diagrammatic)

Following laser wavelengths have been cleared by the Food and Drug Administration (FDA) for tooth whitening.

i. Argon laser (488 nm)
ii. Carbon dioxide (CO_2) (10,600 nm)
iii. Gallium-aluminium-arsenide (GaAlAs) laser (980 nm); diode laser
iv. Photochemical laser whitening (Smartbleach)

i. Argon Laser

Argon lasers are in the form of a blue light with the wavelength of 480 nm in the visible part of the spectrum. Such a spectrum is absorbed by dark color. The affinity to dark stains ensures that the yellow-brown color can be easily removed.

Argon laser rapidly excites the already unstable and reactive hydrogen peroxide (H_2O_2) molecule, followed by absorption of energy into all intramolecular bonds. The H_2O_2 molecules fall apart into extremely reactive ionic fragments that combine with the chromophilic structure of the organic molecules, producing simpler chemical chains; subsequently result in white tooth surface.

Argon lasers emit fairly short wavelengths (480 nm) with higher energy photons; whereas, plasma arc lamps, halogen lamps and other heat lamps emit short wavelengths as well as longer invisible infrared wavelength (750 nm–1.0 μm) with longer energy photons and predictable high thermal character. This high energy can create unfavorable pulpal responses; however, this does not happen in case of argon laser.

ii. Carbon Dioxide (CO_2) Laser

Carbon dioxide (CO_2) lasers having wavelength of 10,600 nm are basically used for enhancing the effect of argon lasers. It is not related to the color of the tooth; and also the energy is emitted in the form of heat. The laser penetrates only 0.1 mm into water and H_2O_2, where it gets absorbed. This energy can enhance the effect of whitening agent after the initial argon laser process. Argon laser, which emits a visible blue light is used first to activate the bleaching gel. This blue light will be absorbed by the dark stains and becomes less effective as the tooth whitens, because the blue light will be reflected rather than absorbed by the whiter tooth surface. Then CO_2 laser, which emits invisible infrared energy is used to achieve deeper penetration of the energized oxygen, resulting in a deeper, more efficient tooth whitening.

Use of carbon dioxide laser has been discontinued because of its thermal effect on the pulp.

iii. Gallium-Aluminum-Arsenide (GaAlAs) Laser; Diode Laser

These are semiconductor lasers with varying wavelengths; 980 nm wavelength is usually used for bleaching procedure.

The surface temperature can be raised from 36°C at 1W to 86°C with 3W. Pulpal temperature increased from 4.3°C to 16°C. The bleaching agent utilizes 35% hydrogen peroxide.

iv. Photochemical Laser Whitening (Smartbleach)

The commonly used bleaching agents (mostly hydrogen peroxide) along with laser tooth whitening systems have certain limitations, such as:

- Commercial hydrogen peroxide system has the potential to affect enamel because of the acidic pH of the solution (5.0 to 6.0).

- The concentrated solutions of hydrogen peroxide (30%) can transiently reduce the microhardness of enamel and dentin.
- Lasers may result in post-treatment sensitivity.

Smartbleach technique (KTP Smartbleach) utilizes alkaline bleaching gel. The primary action of Smartbleaching is photochemical and not photo-thermal. The perhydroxyl radical is produced as compared to superoxide, which is more reactive than the superoxide and other radicals. Also under alkaline conditions, etching of the tooth surface does not occur and the results obtained with KTP and argon lasers are equal to photobleaching.

Smartbleach is preferred choice for bleaching tetracycline-stained teeth. The chelate formed between tetracyclines and hydroxyapatite is a red quinone product dimethylamino tetracycline. This colored product is relatively resistant to oxidation from peroxide, but can be broken down (photo-oxidized) by green light in a particular narrow spectral range (512 to 540 nm). Because this energy aligns particularly well with the wavelength of KTP laser (532 nm), energy from this laser can cause terminal photo-oxidation of the quinone molecule, which renders it colorless. The use of the KTP laser in combination with a hydrogen peroxide-based gel ensures that complete and irreversible bleaching of red quinone occurs.

In addition to driving photo-oxidation reactions within the tooth, some of the visible green laser energy applied to the site is absorbed in Rhodamine B red dye, which is present within the bleaching gel. This is a photochemical process resulting in the production of free oxygen radicals. A portion of the KTP laser energy absorbed into Rhodamine B dye is also transferred from the excited molecule into the bleaching gel in the form of thermal energy. This transfer results in controlled heating of the gel and not the tooth, minimizing the possibility of thermal insult to the pulp. This superficial heating of the gel accelerates the breakdown of hydrogen peroxide, which further boosts the overall yield or free oxygen radicals over a given time.

Figure 32.10A to E depicts laser-assisted whitening of maxillary anterior teeth.

The currently marketed laser whitening systems are as follows:

i. *LaserSmile whitening system (Biolase)*

- A purple gel is applied to the affected teeth at a time; the dental dam gel is also applied to protect the gums.
- Laser light is then applied to the affected teeth for two minutes collectively for adequate whitening.

ii. *PearlinBrite laser whitening system*

- This system utilizes energy transfer crystals (ETCs). ETCs actually absorb laser's energy and transfer the energy to the hydrogen peroxide molecules of the gel.
- This whitens the stains much better than carbamide peroxide gels.
- Usually several applications, lasting up to 30 seconds per tooth is needed per session.

Laser bleaching: Key points

- The studies evaluating bleaching techniques with high concentrations of hydrogen peroxide and with or without the use of light activation resulted in enhanced lightening.
- Only one particular wavelength may perform direct photobleaching (photo-oxidation); example, KTP (532 nm). When KTP is used in combination with a bleaching gel containing a chromophore, allowing the absorption of the laser light, photodynamic reactions can be induced. This combination of wavelength and specifically dyed bleaching gel also allows for safe bleaching (no damage to enamel and pulp).
- A few wavelengths are not recommended for laser bleaching; for example, Nd:YAG, Er:YAG, and CO_2. When using high power diode lasers for bleaching, care has to be taken so as not to overheat the pulp. Also diode lasers are not really advocated for laser bleaching, except when the wavelength is used in combination with a specific wavelength of bleaching gel.
- With the exception of KTP, laser activated bleaching is based on heating of the bleaching gel.
- Long-term evaluations for laser-enhanced bleaching procedures are less documented.

Temperature changes during bleaching

Plasma arc light: The temperature was increased to 39.3°C, when colorant was used and was 37.1°C when no colorant was used.

QTH light: The temperature rise was 24.8°C when colorant was used and 11.5°C when no colorant was used.

Conventional QTH light: It increased temperature by 17.7°C when colorant was used and 11.1°C when no colorant was used.

Argon laser: It increased the temperature by 9.4°C irrespective of the use of colorant.

Diode laser: The increase in temperature was 37°C at 1W and 28.6°C at 3W. The presence of bleaching gel, however, reduces the increased temperature.

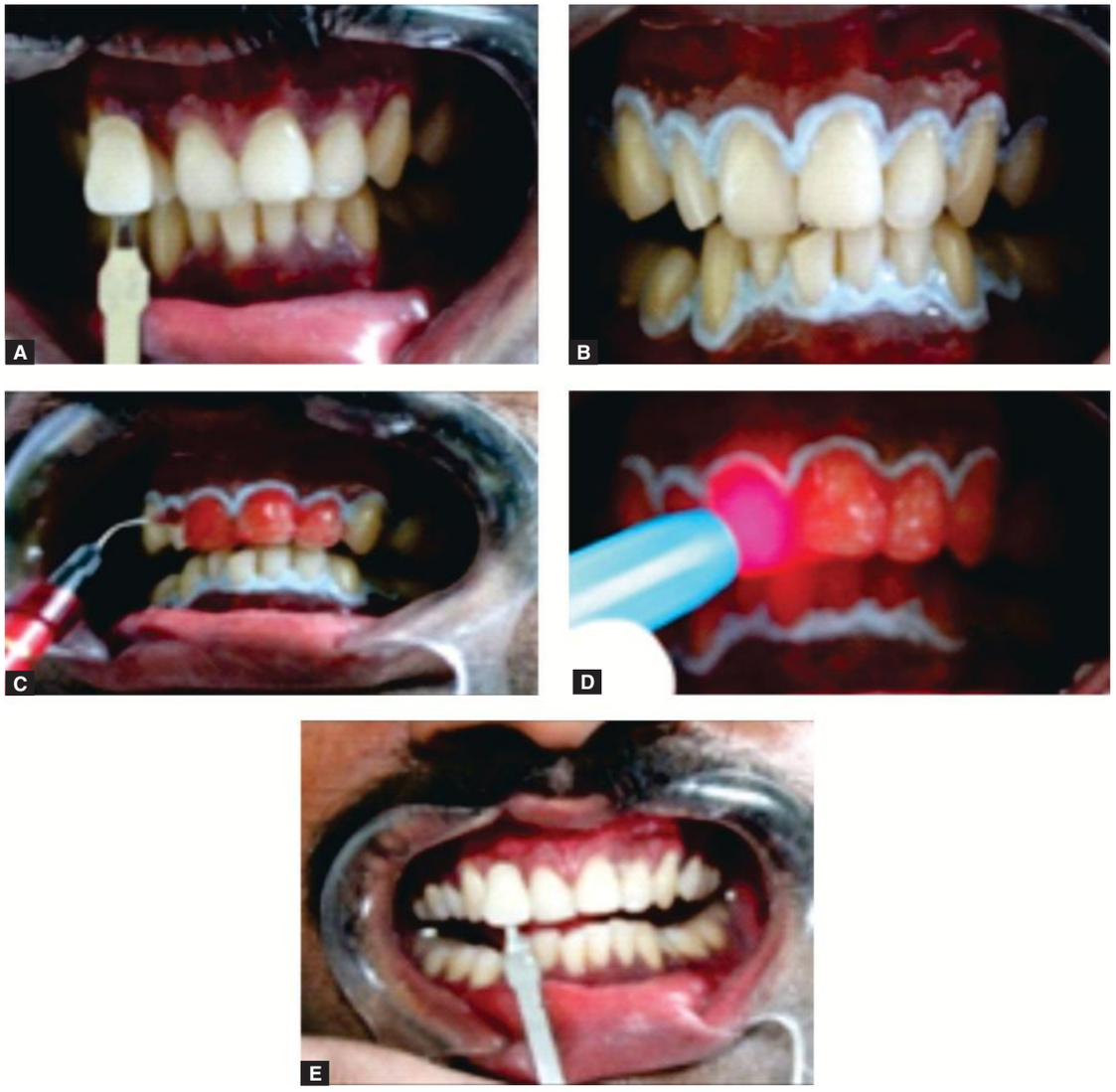


Fig. 32.10: Laser-assisted whitening (maxillary anterior teeth): **(A)** Preoperative; **(B)** Application of liquid dam; **(C)** Application of hydrogen peroxide; **(D)** Laser activation; **(E)** Postoperative

IV. BLEACHING WITH NON-THERMAL ATMOSPHERIC PLASMA

Plasma is considered as the fourth state of matter and it is the most abundant state in the universe. Plasma can be divided into two main categories: Hot plasma (near-equilibrium plasma) and cold plasma (non-equilibrium plasma). Hot plasma consists of very high temperature particles and they are close to the maximal degree of ionization. Cold plasma is composed of low temperature particles and relatively low degree of ionization. Various investigators (Lee, et al 2009) have demonstrated that room temperature plasma could be used for tooth bleaching. The authors opined that the tooth bleaching method with plasma can be complementary to conventional methods because it provides effective bleaching without thermal damage.

V. OVER THE COUNTER PRODUCTS

A variety of tooth whitening products, mainly tooth-polish, toothpastes and bleaching kits are available in the market. Mostly the toothpolish and tooth pastes, contain ingredients that remove extrinsic stains on teeth rather than bleaching, although a few products do contain bleaching agents. Extrinsic stain removal from teeth by toothpastes (Close-up whitening, Amway whitener, Shine and Smile, etc.) is mainly achieved by physical means, though some commercial products claim removal of stains from teeth by chemical means through organic solvents. The readymade bleaching kits supply preformed trays with the bleaching agent dispensed in tubes (Colgate Platinum). The instructions on these kits recommend the use of kit for a particular span of time.

The wearable bleaching devices are as follows:

i. ***Wearable bleaching device:*** The iPower is the first wearable and portable bleaching device. It is safe, powerful, cost-effective, pain-free and easy to use. It uses thermal diffusion technology to deliver optimum results. An accurate amount of thermal energy is delivered onto the bleaching gel, when painted on the patient's teeth. A thermal gradient thus created in the gel increased the diffusion rate of bleaching agent into the enamel structure, speeding up the entire process.

ii. ***Tooth whitening pen:*** It is an easy and convenient way for fast and effective whitening of the teeth. The whitening gel is provided in a pen-like applicator, which is easy to use and carry.
The whitening gel usually contains carbamide peroxide; however, hydrogen peroxide-based pens are also available.
The gel can be selectively applied to individual teeth or just top or bottom of a tooth as needed, e.g. Listerine whitening pen, White ice whitner stick.

Since, over the counter products are not monitored by the professionals, so their potential, efficacy, results and side effects are not thoroughly known. Regarding the bleaching kits, apart from few instructions, no extra knowledge is provided to the patient in relation to the whitening process and exact manner of use. The prefabricated trays may not fit accurately in the patients' oral cavity; may lead to occlusal discrepancies, temporomandibular joint dysfunction and leaching out of the bleaching agent causing soft tissue injury.

VI. ADVANCED TECHNIQUES

i. ***Bleaching technique modified by casein phosphopeptide amorphous calcium phosphate (CPP-ACP):*** Almost all bleaching techniques demineralize enamel; the extent may vary.
The major disadvantage is that the abrasive constituents not only remove the stains but also abrade the tooth structure. This makes the teeth more susceptible to further permeability to extrinsic stains and internalized stains. Remineralization techniques in association with vital tooth bleaching regimens, such as using CPP-ACP based substances during and after bleaching, have been advocated to overcome these problems.

ii. ***Enzymatic bleaching dentifrice:*** Enzymatic dentifrices can be used as bleaching catalyst by generating hydrogen peroxide *in situ*; reducing the risk of toxic effects of hydrogen peroxide, especially when applied in higher concentration.

iii. ***Opalescence (improved):*** A new bleaching agent containing tri-barrel hydremide-peroxide gel with traditional hydrogen peroxide/carbamide peroxide is being tried; however, initial studies have not shown any advantage over traditional bleaching modalities.

ADVERSE EFFECTS OF BLEACHING

The bleaching process, both vital and non-vital, has side effects, adversely influencing the tooth and the surrounding tissues. The adverse effects reported in animal and human studies include:

A. General effects
   a. Hydrogen peroxide toxicity
   b. Effect on restorative materials
B. Effects of vital tooth bleaching
   a. Increased sensitivity
   b. Effect on gingiva and mucous membrane
   c. Effect on enamel
   d. Effect on dentin
   e. Effect on pulp
   f. May activate caries progression
C. Effects of non-vital bleaching
   a. Cervical root resorption
   b. Effect on marginal seal

Adverse effects of bleaching

Vital tooth bleaching
- Increased sensitivity
- Alteration in enamel structure
- Minor ulceration of gingiva
- Affects color of dentin
- Affects adhesion
- Reduction in pulpal blood circulation
- Caries progression

Non-vital tooth bleaching
- Cervical root resorption
- Effect on marginal seal (enhance microleakage, especially in composites).

General effects
- Hydrogen peroxide toxicity (toxicity of bleaching agents), viz. skin burns, tissue emphysema, etc.
- Effect on restorative materials, such as increase in solubility, decrease in bond strength, etc.

A. General Effects

a. Hydrogen Peroxide Toxicity (Toxicity of Bleaching Agents)

Hydrogen peroxide, the mainly used bleaching agent, is a toxic product, especially at higher concentrations. The adverse effects of hydrogen peroxide are:

i. Hydrogen peroxide can cause burns on contact with skin or eyes. When 30% H_2O_2 was applied to the tip of rat tongue at 15-minute intervals (four

applications), it led to edema. However, prior application of catalase may prevent such edema.

ii. The inadvertent flow of hydrogen peroxide into the periodontal ligament and periapical tissues during root canal treatment, produce effervescence and liberate oxygen; may lead to tissue emphysema.

iii. The adverse effects of hydrogen peroxide mouth rinses (6% H_2O_2) include irritation of oral mucosa and discomfort, dryness, loss of taste, elongation of filiform papillae and diffuse mucosal whitening of tongue.

iv. Use of hydrogen peroxide may lead to change in epithelial morphology and its proliferation rate. The PCNA (proliferating cell nuclear antigen) index, an indication of cell proliferation, increases in basal and parabasal layers of epithelium. Usually smokers had a significantly higher PCNA index than non-smokers; this difference minimizes following bleaching, indicating stimulation of cell division activity by peroxide similar to that produced by smoke. It has been established that 10% carbamide peroxide would act as a tumor promoter in the presence of mutated cells.

v. The pulp is reported to have a low peroxidase enzyme activity due to a sparse population of fibroblasts. The inhibition as well as inactivation of pulpal enzymes by hydrogen peroxide has been established. The quantity of peroxide penetrating the pulp chamber, even in small amounts, are sufficient to produce toxic effects on fibroblasts.

b. *Effect on Restorative Materials*

Bleaching may increase the solubility of glass-ionomer and other cements and reduce the bond strength between enamel and resin-based fillings, especially in the first 24 hours, but not later. Following bleaching, hydrogen peroxide residues in the enamel may inhibit the polymerization of resin-based materials and reduce bond strength. Therefore, tooth-bleaching agents should not be used at least 24 hours prior to treatment with resin-based materials.

To improve the bond strength of previously bleached teeth, 10% sodium ascorbate has been found to be effective in reversing the compromised bonding of composite with enamel, previously bleached with 10% carbamide peroxide.

It has been established that sodium ascorbate gel when applied following bleaching increased the resin–enamel bond strength. (The increase was proportional to the duration of application.)

A few authors opined that use of catalase or catalase-like substances can effectively neutralize the residual hydrogen peroxide on the bleached teeth; tocopherol and grape seed extract have also been found to be effective. It is recommended to postpone the bonding procedure for a sufficient time after the bleaching; 24 hours to 4 weeks have been quoted in various studies.

B. Effects of Vital Tooth Bleaching

a. *Increased Sensitivity*

Vital tooth bleaching uses high concentration of hydrogen peroxide (35–50%) or carbamide peroxide (35–40%), often supplemented with a heat source; or in a bleaching tray to be used at home.

The exact mechanism of tooth sensitivity is not clear; however, on the basis of *in vitro* studies, it is assumed that H_2O_2 in the bleaching gel is capable of penetrating through enamel and dentin and reaching the pulp chamber.

It has been established that the bleaching procedure increases intrapulpal temperature, which is detrimental to pulp. The heat when applied to hydrogen peroxide may lead to diffusion of H_2O_2 into dentin and pulp. However, inflammatory changes get recovered within one month.

In-office bleaching procedures with high concentrations of hydrogen peroxide were associated with high post-treatment sensitivity; however, night-guard bleaching with lower concentration of hydrogen peroxide (3.0%) or carbamide peroxide (10%) has resulted in decreased sensitivity.

The pH of the bleaching agent and desiccation of the tooth surface are causative agents for sensitivity. Potassium nitrate and sodium fluoride when added to 10% carbamide peroxide, considerably reduce sensitivity without any significant change in bleaching results. Use of anti-inflammatory drugs prior to bleaching may also be effective.

Tooth sensitivity is a common adverse effect of external tooth bleaching. Various studies have confirmed that use of 10% carbamide peroxide had led to increased tooth sensitivity in at least 50% of the patients. Higher incidences of tooth sensitivity (65–78%) have been reported after bleaching with H_2O_2 in combination with heat.

Tooth bleaching in younger individuals (less than 18 years) should be avoided; and also on teeth with caries, exposed dentin and/or defective restorations.

b. *Effect on Gingiva and Mucous Membrane*

Bleaching trays which are ill-fitting, overextended, or not properly trimmed, can irritate soft tissues directly or through leaking bleaching agents. Higher concentration of H_2O_2 produce soft tissue burns; the color may change to white. It is hypothesized that

rehydration and application of antiseptic ointment return the color to its original. Effective protection of gingiva and mucosa, with good isolation will minimize the risk of irritation. The barrier must be checked for any sign of leakage. The barrier should cover all buccal gingival surface.

In case of ulceration, the site is extensively washed with water and/or the anesthetic solution. In severe cases, vitamin E can be applied over the ulceration, which will help healing the wound. The position of light should also be adjusted; should not be too near to mucous membrane.

c. *Effect on Enamel*

Vital tooth bleaching using carbamide peroxide or hydrogen peroxide significantly alters enamel topography. High concentrations of carbamide peroxide damage enamel surface integrity; however, lesser than phosphoric acid etch. As a result of this increased surface roughness, the teeth may be more susceptible to extrinsic discoloration after bleaching. A few studies have reported that after bleaching, enamel gets weakened by oxidation of organic/inorganic elements, which may appear as porous or pitted surface. The oxygen radicals released from peroxide bleaching material may affect chemical structure of enamel, reducing calcium/phosphate ratio, enamel microhardness and surface morphology; subsequently, lead to increased caries risk. Various studies have also confirmed that use of 10% carbamide peroxide daily for 6 months did not adversely affect the surface texture of human enamel.

d. *Effect on Dentin*

Bleaching agents affect tooth color (primarily of dentin).

McCaslin et al (1999) placed 10% carbamide peroxide directly on enamel and observed a uniform color change throughout the dentin surface. The majority of studies investigating surface morphology of dentin found no significant changes. However, Zalkind et al (1996) found that Ca and PO_4 ions ratio of dentin was modified following 7 days of continuous treatment with 30% H_2O_2 or 10% carbamide peroxide solutions.

A transient decrease in surface microhardness has been observed; however, it is recovered following treatment with fluoride solutions. It is established that oxygen radicals damage the dentin substrate (hybrid layer), which is mainly responsible for adhesion of composite resin to dentin surface. Recent studies have also confirmed that the durability of adhesive restorations are detrimentally influenced by carbamide peroxide bleaching. It is concluded that the damage caused to restoration–dental tissue bond strength by the bleaching agent is proportional to the increase in carbamide peroxide concentration; the dentin–composite bond was less sensitive as compared to enamel–composite bond.

e. *Effect on Pulp*

A 3.0% solution of hydrogen peroxide is capable of causing transient reduction in pulpal blood circulation and occlusion of pulpal blood vessels. The most common side effect experienced by patients using home bleaching technique is mild and transient sensitivity. This sensitivity appears to be dose related rather than pH related.

f. *Effect on Caries Progression*

It is established that vital bleaching alters surface enamel structure (pitting, etc.). Such altered enamel surface, because of bleaching, may activate caries progression.

C. Effects of Non-Vital Tooth Bleaching

a. *Cervical Root Resorption*

The most common complication of non-vital bleaching is cervical root resorption (an inflammatory mediated external resorption of the root). The incidence of cervical root resorption after non-vital bleaching ranges from 5 to 10%. This is more commonly seen in teeth, which become pulpless at a young age and when no barrier is placed between endodontic obturating material and pulp chamber during bleaching. In such cases, the dentinal tubules remain patent and communicate with periodontal space through the defects of cementoenamel junction. This allows the bleach solution to reach the periodontal ligament from root canal system and an inflammatory reaction can be initiated, resulting in external cervical root resorption.

The features affecting the root resorption are:

- 10% of anterior teeth have cervical areas in which enamel and cementum do not meet.
- Cervical resorption occurs coronal to the endodontic seal.

The critical region for the root resorption is the proximal area, where the cemento-enamel junction dips, facilitating cervical root resorption. Location/shape of cervical area and the barrier between endodontic obturating material and the pulp chamber effectively minimizes root resorption.

The internal level of barrier should be one millimetre incisal to the corresponding external

probing of the epithelial attachment. The intention of barrier placement is to cover dentinal tubules apical to the epithelial attachment so that the bleaching agents are contained within the pulp chamber. Cavit and light cure glass-ionomer cements are the most commonly used barrier materials.

b. *Effect on Marginal Seal*

The non-vital bleaching procedures adversely affect the marginal seal leading to marginal leakage. Various studies have concluded that use of mouthrinses (Oral-B, Listerine, etc.) after bleaching can increase post-restoration microleakage of composite restorations.

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